LAB-2 WORKSHEET (31ST January)

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GITHUB LINK - https://github.com/Layaa-V/MLOps-LayaaVishwakarma-M25CSA017/tree/Lab-2-Worksheet
WANDB LINK - https://wandb.ai/vishwakarmalayaa-/cifar10-assignment-lab2?nw=nwuservishwakarmalayaa

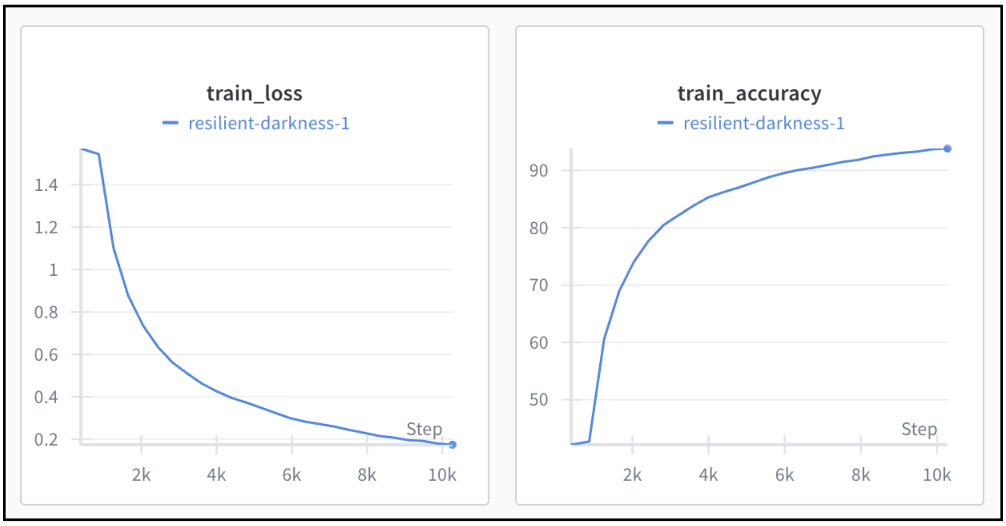
OVERVIEW -
In this experiment trained a Convolutional Neural Network (CNN) on the CIFAR-10 dataset, analyzed its computational complexity (FLOPs), and visualize the internal training dynamics (Gradient Flow and Weight Updates) using a custom training loop and WandB integration.

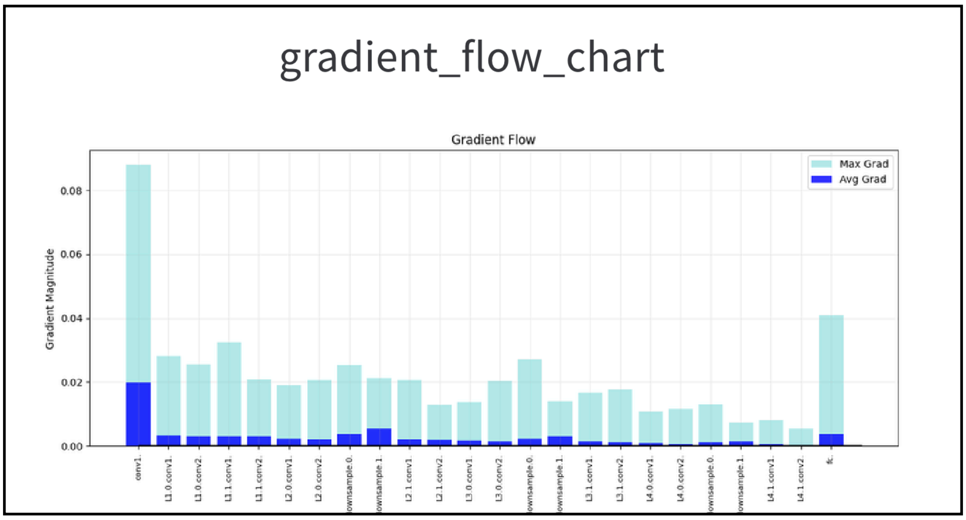
KEY NOTES IN IMPLEMENTATION-
1. Instead of a standard CNN, I utilized a ResNet-18 architecture. To make it compatible with the smaller CIFAR-10 image size (32 x 32), the following modifications were made:
   - **Input Layer:** The initial 7x7 convolution (stride 2) was replaced with a 3x3 convolution (stride 1). This prevents aggressive downsampling that would destroy features in small images.
   - **Pooling:** The initial MaxPool layer was removed to preserve spatial resolution in early layers.
   - **Output Layer:** The final Fully Connected (FC) layer was modified to output 10 classes instead of 1000.

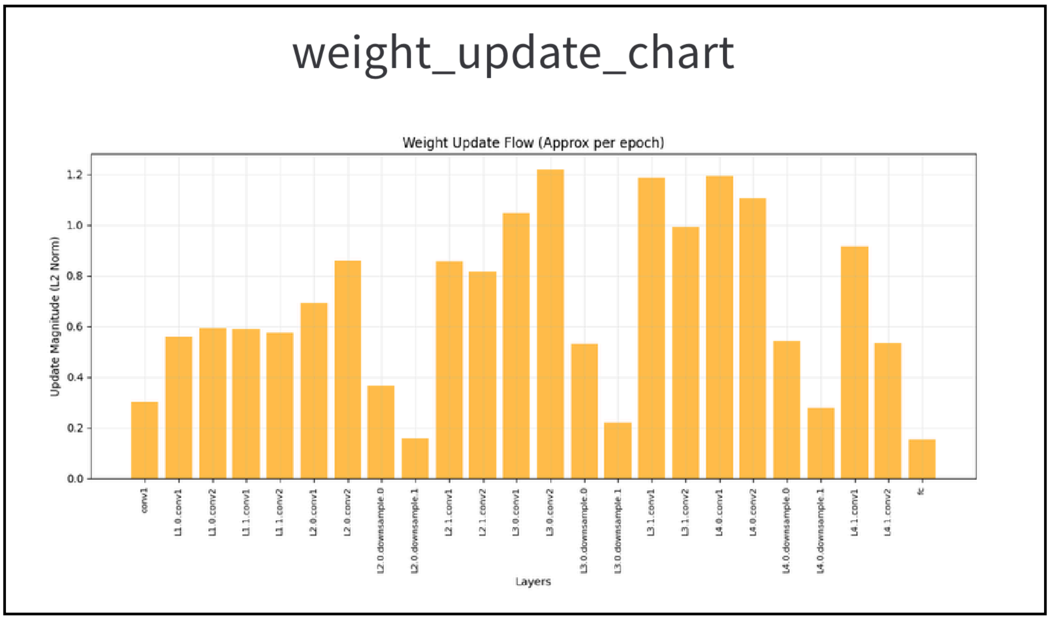
2. Custom DataLoader was implemented.
3. Using the 'thop' library, the theoretical floating-point operations (FLOPs) were calculated. (Result being 557.889M)

TRAINING RESULTS -
Hyperparameters:
- Epochs: 25
- Batch Size: 128

Performance:
- Final Training Loss: 0.17384
- Final Accuracy: 93.824







The model showed consistent convergence, with accuracy improving steadily over the 25 epochs. The use of data augmentation helped bridge the gap between training and validation performance.